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Network Management

## 1.

In a company network, a router connects different departments to the internet. One day, the company decides to improve security and update the router configuration. The network administrator first backs up the current configuration to avoid losing data. Then the administrator logs into the router using secure remote access such as SSH.

The update includes changing the IP addressing scheme, enabling a firewall rule, and updating the routing table. The administrator carefully enters the new configuration commands and tests connectivity between departments. After that, the internet connection is tested to make sure users can still access external services.

If everything works correctly, the configuration is saved permanently. If problems occur, the administrator restores the previous backup. This process ensures the network remains stable, secure, and available. Proper planning, testing, and backup are important steps in any router configuration update.

## 2.

Ansible is an automation tool used for configuration management, application deployment, and network management. It helps administrators automate repetitive tasks instead of manually configuring each device.

Ansible uses simple text files called playbooks written in YAML format. These playbooks describe the tasks that need to be performed, such as installing software, updating configurations, or restarting services. Ansible connects to devices using SSH and does not require special software to be installed on client machines.

In network management, Ansible can configure multiple routers and switches at the same time. This reduces human error and saves time. Ansible improves consistency, efficiency, and control in managing modern networks.

## 3.

In GNS3, a network simulation can be created using routers and switches. To simulate a backup, I first created a simple topology with one router and two PCs. After configuring the router with IP addresses and routing rules, I saved the running configuration.

Using the router command line interface, I copied the running configuration to a startup configuration file. This acts as a backup in case the router restarts. I also exported the project file in GNS3 to keep a full backup of the network setup.

From this practice, I learned that backups are very important in network management. If a configuration error happens, it is easy to restore the previous settings. Simulating backups in GNS3 helps understand real-world network protection and recovery processes.

## 4.

In Draw.io, the configuration process flowchart starts with identifying the need for a network change. The next step is planning the configuration update. After planning, a backup of the current configuration is created.

The next box shows implementing the configuration changes on the router or device. After implementation, the system is tested to check connectivity and performance. A decision symbol can be added to ask if the configuration works correctly.

If the answer is yes, the configuration is saved permanently. If the answer is no, the backup is restored. This flowchart clearly shows the logical steps in a network configuration process and highlights the importance of testing and backup.